VERITAS: Multimodal Document Intelligence

Sample Research Document for PDF Extraction Testing

Abstract

VERITAS is a prototype document intelligence platform designed to extract, organize, and retrieve information from digital documents. It combines document processing, semantic embeddings, vector search, and evidence-grounded question answering. This sample file is provided to test page-wise PDF text extraction.

1. Introduction

Organizations store valuable knowledge in reports, manuals, research papers, and scanned records. Finding a precise answer in a large collection of documents can be difficult when users do not know the exact keywords used in the source. A document intelligence system can help by extracting text and preserving its relationship to the original page.

2. Document Processing

A PDF extraction pipeline opens a document, visits each page, and retrieves selectable text. The extracted output should retain the original page number. Pages containing only scanned images may return little or no text and require an optical character recognition (OCR) stage.

3. Semantic Embeddings

An embedding model converts a piece of text into a numerical vector. Texts with related meanings can have vectors that are close under a similarity measure. VERITAS uses a sentence-transformer model to generate 384-dimensional vectors for text chunks.

For example, the question 'How can I find useful information?' may be related to a source sentence such as 'The system retrieves relevant passages using semantic search.' The wording differs, but the underlying meaning is similar.

4. Vector Retrieval

A vector database extension can compare a query embedding with stored chunk embeddings. Cosine distance is one common measure. Results can be ordered from the smallest distance to the largest distance to retrieve the nearest vectors.

5. Evidence-Grounded Answers

A retrieval-augmented question-answering workflow first retrieves relevant document passages. A language model can then draft an answer using those passages as evidence. To support verification, each passage should retain metadata such as its document identifier and source page number.

6. Conclusion

A reliable document intelligence pipeline should preserve source information through every processing stage. Page-aware extraction, chunking, embeddings, and vector retrieval form important foundations for evidence-grounded answers. This sample document contains three pages so that page numbering can be checked during development.